Permutation Chi-square Test via Monte Carlo Simulation is a non-parametric statistical method for assessing whether the distribution of a categorical variable is consistent across multiple independent groups. Its core logic relies on constructing an empirical distribution of the test statistic under the null hypothesis through large-scale random permutations, thereby circumventing the reliance of the traditional chi-square test on asymptotic assumptions regarding sample size and expected frequencies.

Specifically, suppose there are G independent groups ($G \geq 2$), with sample size n_i for the i-th group and total sample size N

$$N = \sum_{i=1}^{G} n_i \tag{1}$$

Let X be a categorical variable with K categories; the data can be organized into a $G \times K$ contingency table, where O_{ij} denotes the observed frequency in the i-th group and the j-th category. Under the null hypothesis H_0 that the distribution of X is independent of the grouping variable Y(i.e., no significant difference in category proportions across groups), the Pearson chi-square statistic χ^2_{obs}

$$\chi^2_{obs} = \sum_{i=1}^{G} \sum_{j=1}^{K} \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}} \tag{2}$$

is used as a measure of observed discrepancy, where the expected frequency E_{ij} is computed from the row and column margins of the table:

$$E_{ij} = \frac{\left(\sum_{j=1}^{K} O_{ij}\right) \times \left(\sum_{i=1}^{G} O_{ij}\right)}{N} \tag{3}$$

To construct the empirical distribution of this statistic under the null, a non-parametric permutation procedure is employed. The algorithm proceeds as follows:

First, all categorical observations from the G groups are pooled into a single vector X_{pooled} of length N; Then, in each permutation, the sample sizes $n_1, n_2, \ldots, n_G$ are kept fixed (i.e., the original group structure is preserved), but the order of the observations in X_{pooled} is completely randomly shuffled; The shuffled vector is then reassigned to the G groups according to the original sample sizes, creating a new "pseudo-contingency table" for which the chi-square statistic $\chi_{(b)}^2$ is computed, where $b = 1, 2, \ldots, B$ (B is the total number of permutations, set to 10,000 in this study). Repeating this process B times yields the empirical distribution $\{\chi_{(1)}^2, \chi_{(2)}^2, \ldots, \chi_{(B)}^2\}$ of the permutation statistics, which precisely reflects the range of chi-square values attributable to random fluctuations alone when H_0 is true.

Based on this distribution, an empirical p-value is used for statistical decision-making. The p-value is defined as the proportion of permutation statistics that are greater than or equal to the observed statistic χ_{obs}^2. To avoid the extreme case of $p = 0$ due to a finite number of permutations,

the corrected formula recommended by Phipson and Smyth is adopted:

$$p = \frac{\sum_{b=1}^{B} I\left(\chi_{(b)}^2 \geq \chi_{obs}^2\right) + 1}{B + 1} \tag{4}$$

Where $I(\cdot)$ is an indicator function (equal to 1 if the condition holds and 0 otherwise). This correction adds 1 to both numerator and denominator, effectively controlling the Type I error rate. In the implementation, special attention must be given to the issue of uniform category coding: the original category labels from all groups (whether numeric or string) should be merged and globally encoded in a unified manner to ensure that identical categories across different groups are assigned the same numeric index, thus constructing the correct $G \times K$ contingency table. Concurrently, the code employs a conditional permutation strategy, i.e., fixing the marginal distribution of group sample sizes (row sums) and only shuffling the sequence of the categorical variable X. This strategy is equivalent to randomly permuting the distributions of the G populations while preserving the sampling error structure, which fully adheres to the basic assumption of homogeneity testing in contingency tables. Ultimately, the magnitude of the observed chi-square value χ_{obs}^2 reflects the actual degree of difference in category composition among groups, with larger values indicating more pronounced group differences. The corresponding p-value is used for statistical inference: if $p < 0.05$, H_0 is rejected, indicating a statistically significant difference in the distribution of the categorical variable across groups (i.e., baseline imbalance); otherwise, the groups are considered comparable (baseline balanced).

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